HealthConnect Clinic

Week 5 — Data Analytics Report

Reducing Missed Appointments Through Data-Driven Insights

AnalystLab Africa Experience Lab

Track: Data Analytics | Week 5

Project question: How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?

Part 1 — Week 4 Foundation (Brief Recap)

This section summarises the Week 4 foundation this report builds on. Replace with your actual Week 4 submission content before final submission.

Problem defined: HealthConnect Clinic experiences a high rate of missed appointments (no-shows), reducing clinical capacity utilisation, lengthening waiting lists, and limiting patient access to care.

Week 4 approach: review the appointment dataset and data dictionary, assess data quality, identify candidate business questions, and shortlist potential KPIs.

Resources used: HealthConnect_Appointment_Data.csv, HealthConnect_Data_Dictionary.xlsx, and HealthConnect_Clinic_Knowledge_Base.docx (for cross-track context).

Key assumptions/limitations carried forward: the dataset is fictional/synthetic; Cancelled appointments are treated as distinct from No-Shows; no patient-identifiable information is present.

Proposed Week 5 focus: move from planning into data preparation, EDA, KPI calculation, an initial dashboard, and business insights.

Part 2 — Data Preparation

The HealthConnect_Data_Dictionary.xlsx was used to confirm expected data types, categories, and value ranges for all 18 fields before analysis. The original CSV file was not modified; all preparation was performed on an in-memory working copy (5,000 rows).

Data Quality Findings

- Data types: all columns loaded with correct types (dates, integers, decimals, text) with zero coercion errors — the file matches the data dictionary exactly.
- Missing values: reminder_channel is missing for 1,366 rows — this is structural, not an error, since every one of those rows has reminder_sent = "No". distance_to_clinic_km is missing for 90 rows (1.8%) and waiting_time_minutes for 60 rows (1.2%); both left as-is and excluded row-wise per analysis rather than imputed.
- Duplicates: 0 fully duplicated rows and 0 duplicate appointment_id values — the primary key is intact.
- Logical consistency: 0 rows where previous_no_shows exceeds previous_appointments; 0 mismatches between booking_lead_days and the calculated date difference; 0 mismatches between appointment_day and the actual weekday; 0 mismatches between age and age_group.
- Category values: all categorical fields (gender, appointment_type, reminder_channel, appointment_outcome, etc.) match the data dictionary's expected categories with no typos or inconsistent labels.
- Ranges: age 18–80, distance 0.5–45km, waiting time 2–68 minutes, booking lead time 0–60 days — all plausible for an outpatient clinic, with no negative values or impossible outliers.

Conclusion: the dataset required no corrective cleaning beyond documenting the above; it was analysis-ready as provided.

Part 3 — Exploratory Data Analysis

Appointment attendance was investigated against appointment characteristics, patient characteristics, appointment history, reminders, waiting time, and distance to the clinic.

Overall Attendance Picture

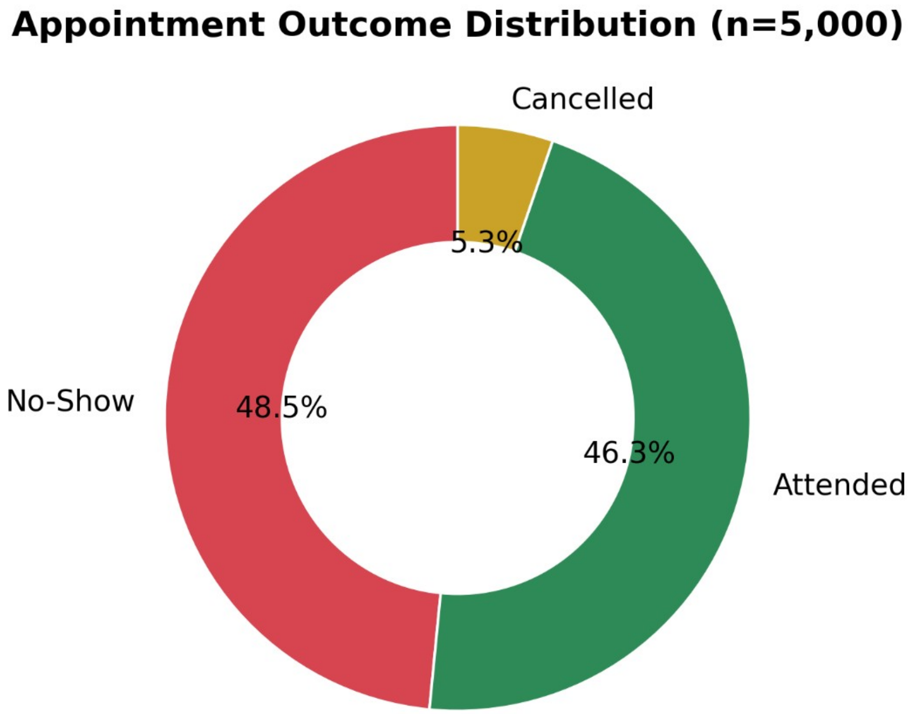

Figure 1. Appointment outcome distribution across all 5,000 appointments.

Almost half of all scheduled appointments (48.5%) end in a No-Show — outnumbering Attended appointments (46.3%) — with 5.3% Cancelled in advance. Missed appointments are HealthConnect's single largest attendance problem, larger than the cancellation rate.

Booking Lead Time

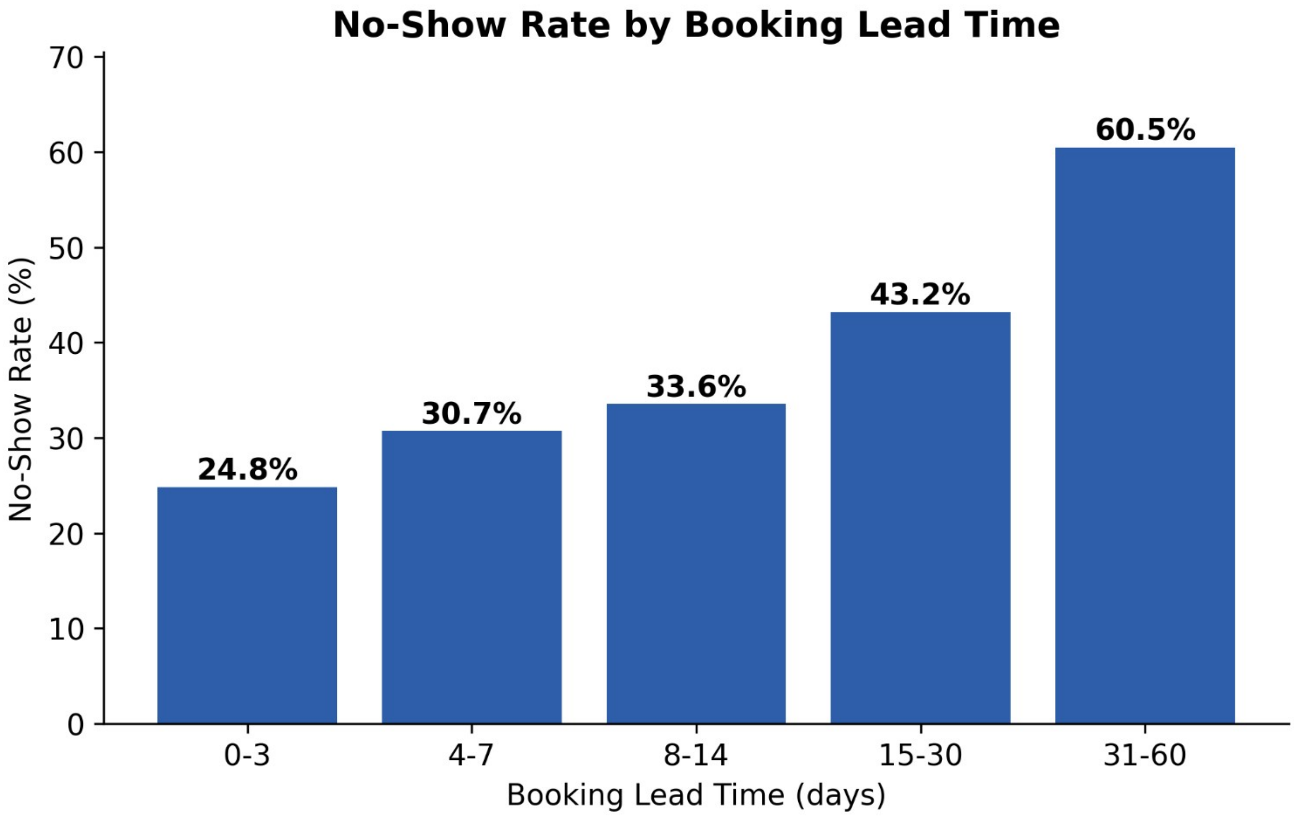

Figure 2. No-show rate by number of days between booking and appointment.

This is the strongest relationship found in the data. No-show rate rises from 24.8% for appointments booked 0–3 days ahead to 60.5% for appointments booked 31–60 days ahead — a 35.6 percentage-point gap.

Distance to Clinic

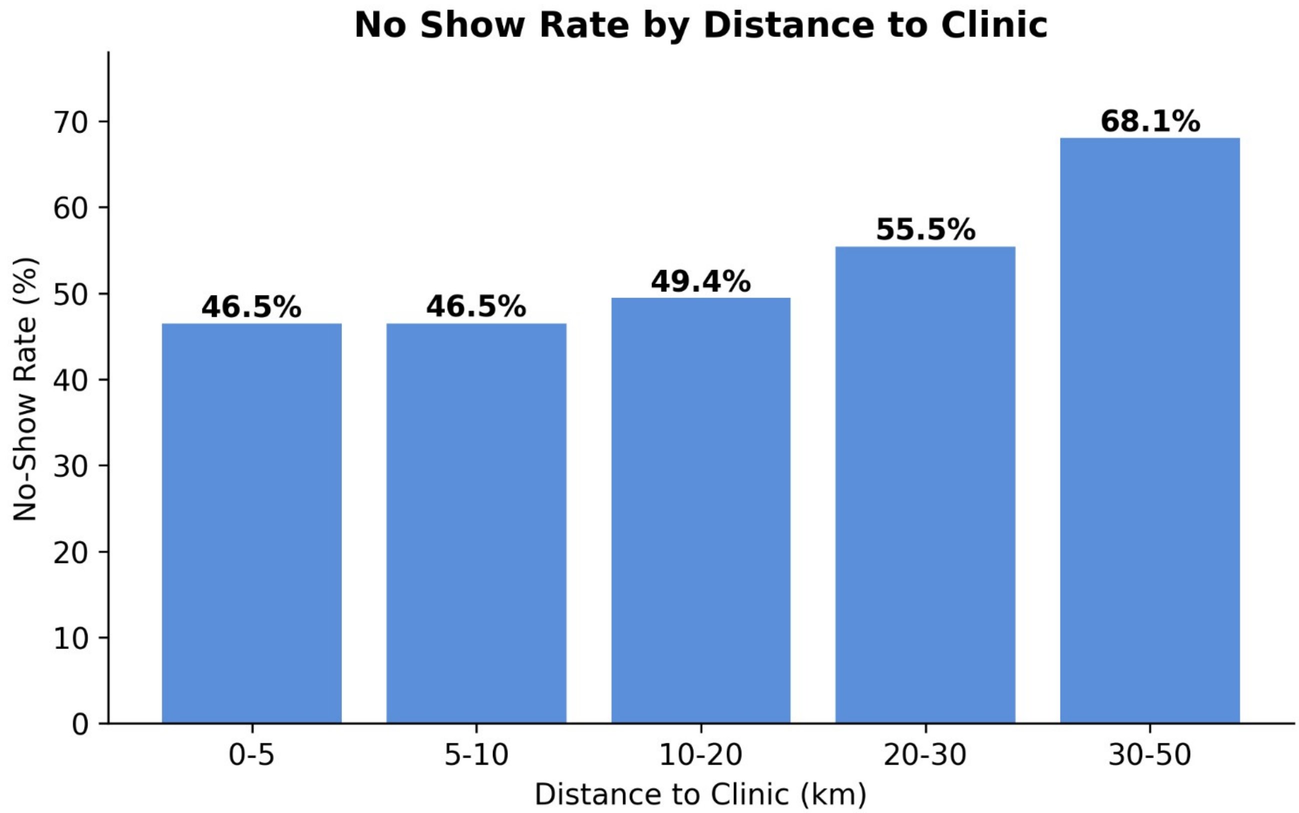

Figure 3. No-show rate by patient distance from the clinic.

No-show rate climbs from 46.5% for patients within 5km to 68.1% for patients 30–50km away, indicating a real travel/access barrier independent of reminders.

Prior No-Show History

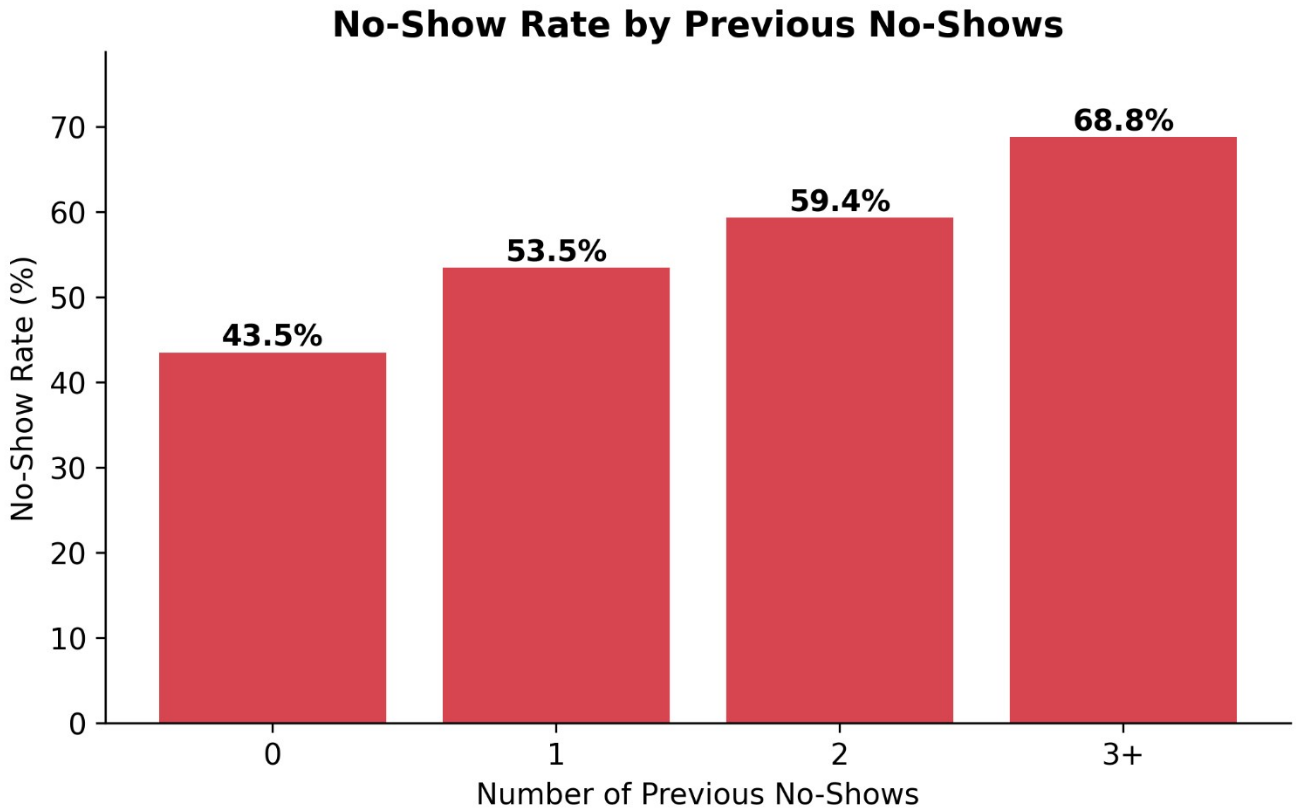

Figure 4. No-show rate by number of previous no-shows on record.

Patients with 3+ prior no-shows miss 68.8% of appointments versus 43.5% for patients with a clean record — past behaviour is a strong, readily available predictor of future behaviour.

Reminders

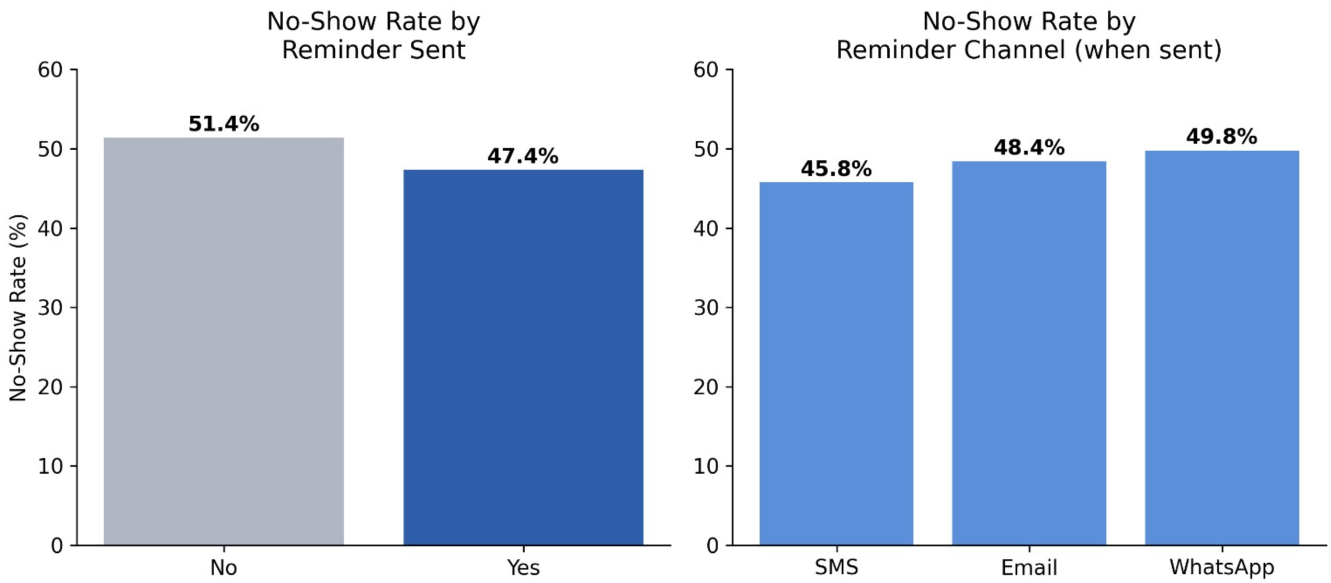

Figure 5. No-show rate by whether a reminder was sent, and by reminder channel.

Sending a reminder reduces no-shows from 51.4% to 47.4% — a real but modest 4.0 percentage-point improvement. SMS is the best-performing channel (45.8%), ahead of Email (48.4%) and WhatsApp (49.8%).

Waiting Time and Other Factors

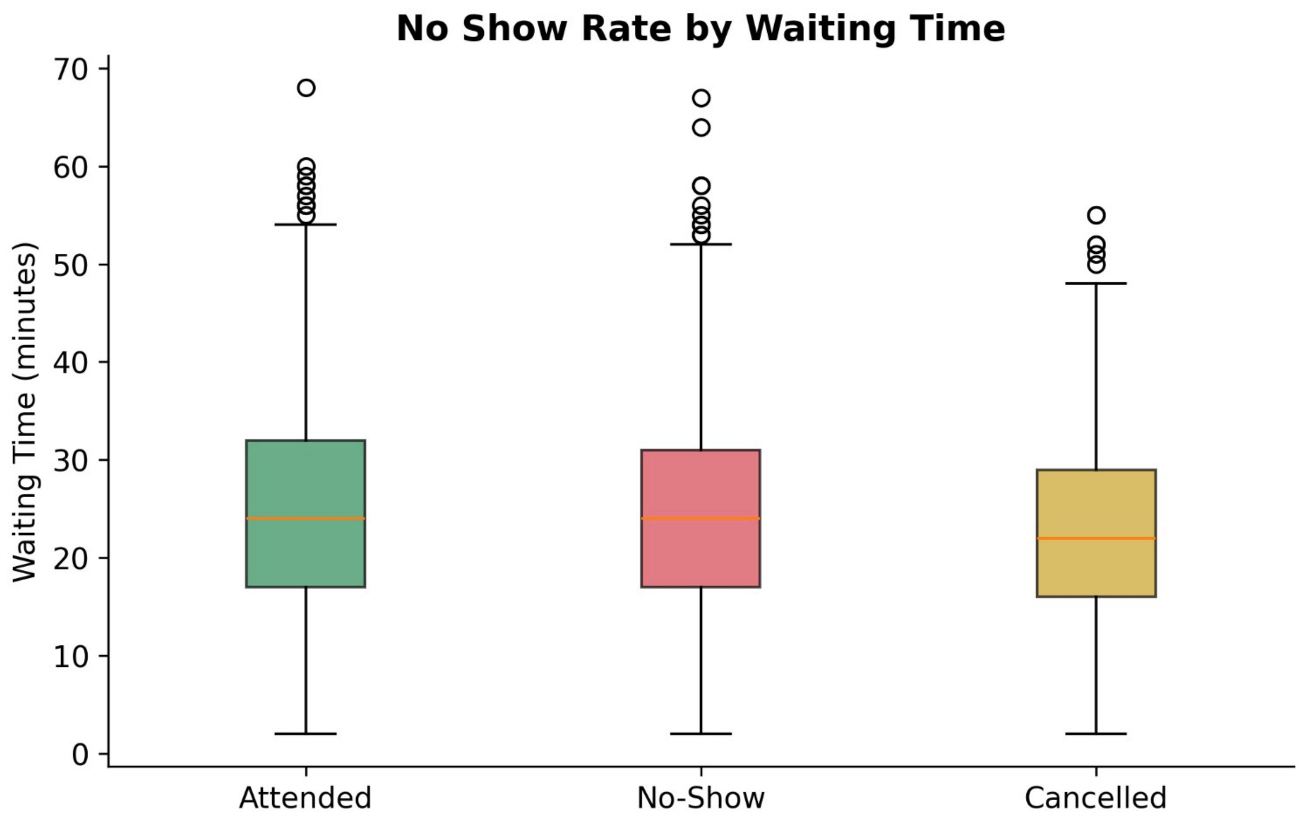

Figure 6. Waiting time distribution by appointment outcome.

Average waiting time is nearly identical across Attended (24.3 min), No-Show (24.2 min) and Cancelled (23.2 min) appointments — waiting time is not a driver of no-shows. Appointment type, time of day, day of week, and age group similarly showed no-show rates clustered tightly around the 46–51% overall average, with no group standing out as a meaningful driver on its own.

Confirming the Lead-Time Effect Across Appointment Types

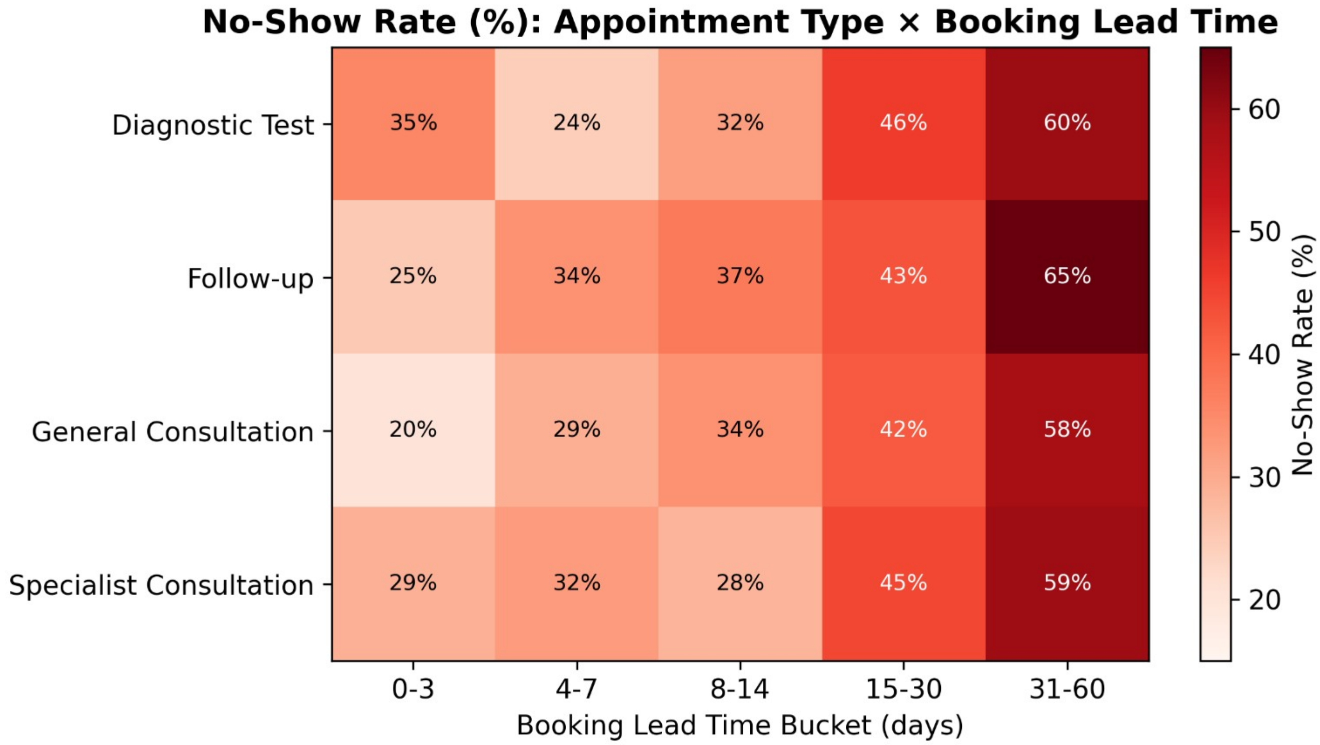

Figure 7. No-show rate (%) by appointment type and booking lead-time bucket.

The lead-time effect holds consistently across every appointment type, confirming it is a genuine attendance driver rather than an artefact of one specific service line.

Appointment Day

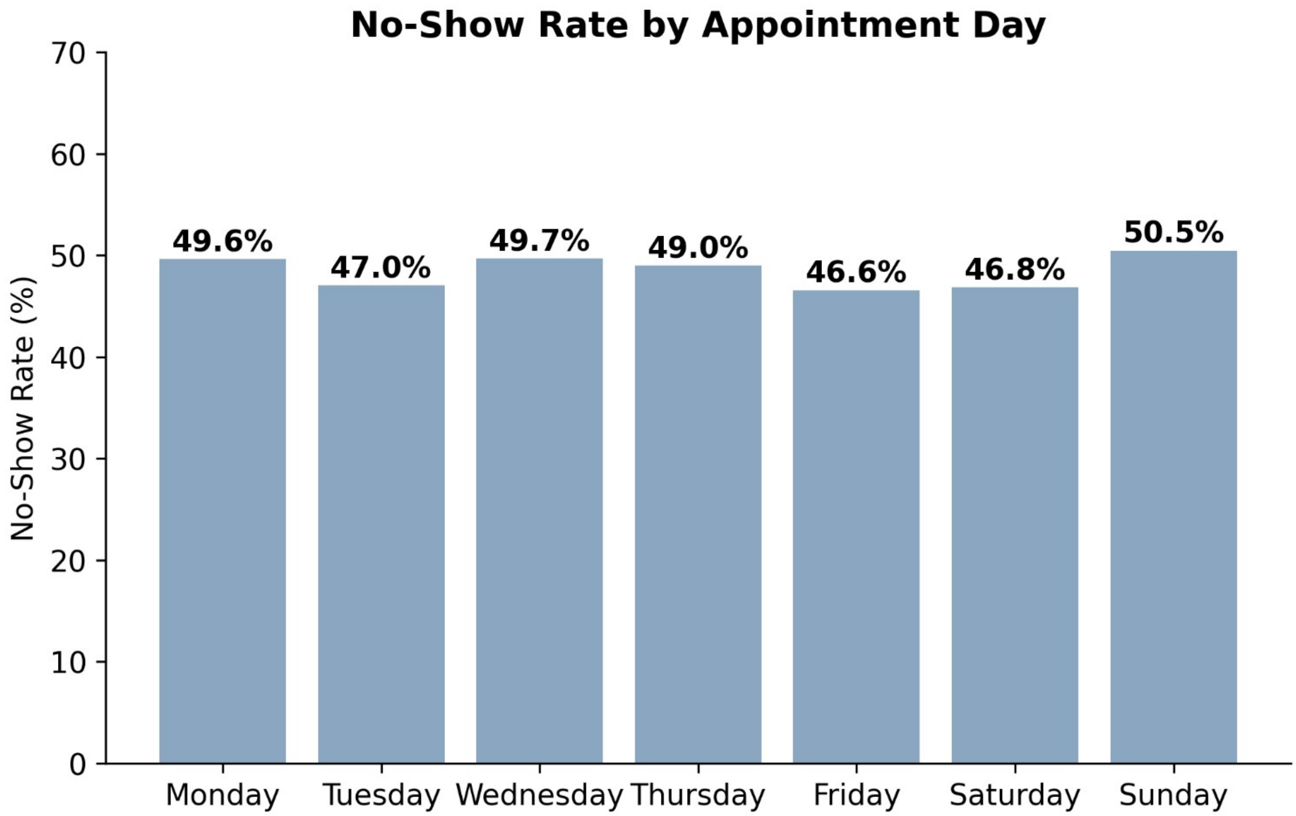

Figure 8. No-show rate (%) by appointment day

Appointment day Sunday has the highest no-show rate (50.5%) and Friday the lowest (46.6%). Day-level differences are present but smaller than the patterns associated with lead time and previous no-shows.

Part 4 — KPI Development

Five KPIs were selected from the Week 4 candidate list, each defined, calculated, and tied to a business question.

KPI	RESULT	WHY IT MATTERS
Total Appointment	5000	Measures Appointment Volume
Attendance Rate	46.28%	Tracks Successful Attendance
No Show Rate	48.46%	Core Missed-Appointment Indicator
Cancellation Rate	5.26%	Separates Cancellations From Missed Attendance
No-Show Rate Among Non-Cancelled	51.15%	Focuses Attendance Problem Excluding Cancellations

Part 5 — Analytical Dashboard Summary

The four strongest attendance drivers are consolidated below into a single at-a-glance dashboard view for stakeholders.

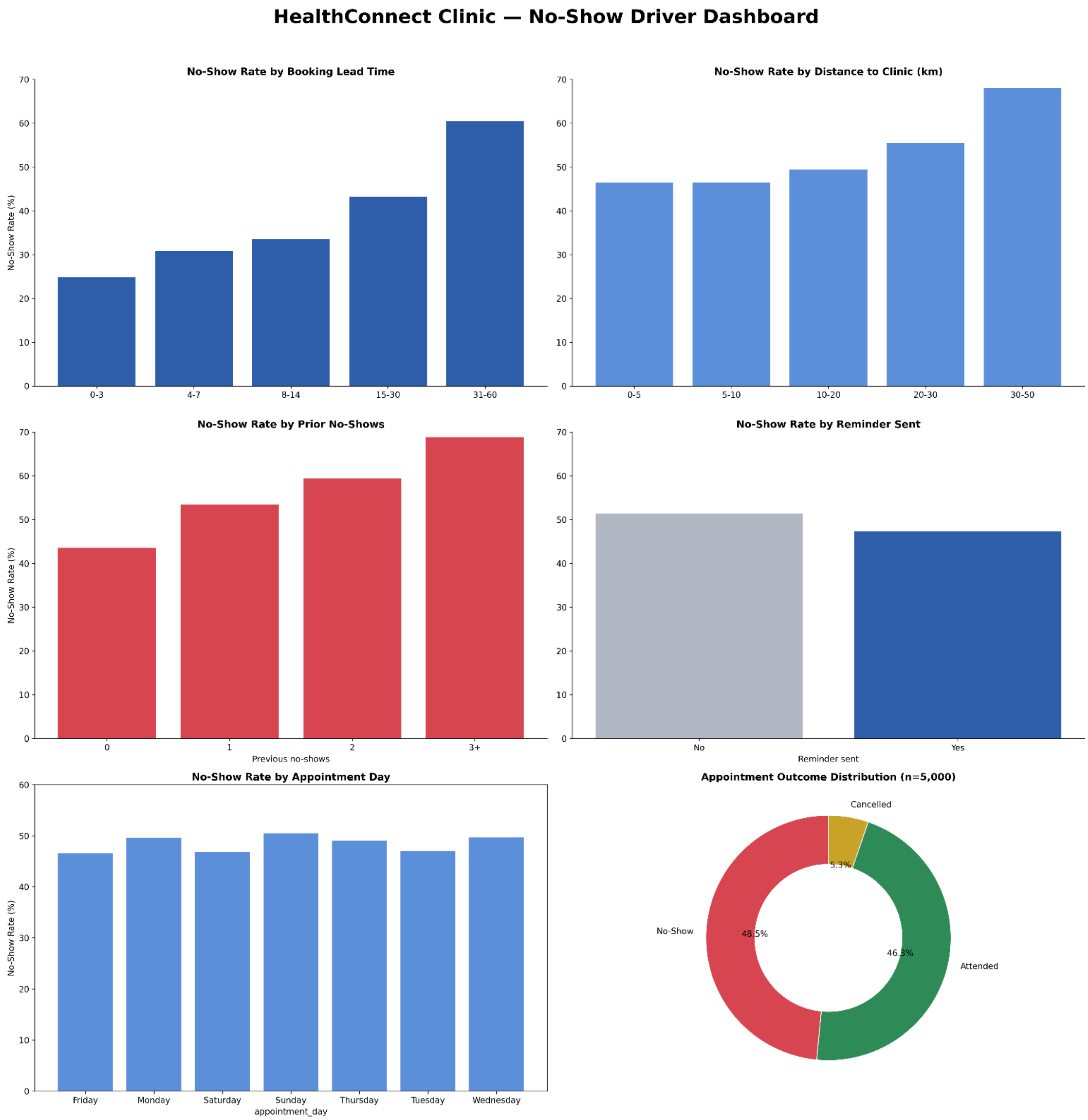

Figure 8. HealthConnect Clinic — No-Show driver dashboard.

Part 6 — Business Insights & Recommendations

Key Insights

- Missed appointments are HealthConnect's attendance problem — at 48.5%, No-Shows outnumber Attended appointments, well ahead of the 5.3% Cancellation rate.

- Booking lead time is the strongest driver of no-shows, rising from 24.8% (0–3 days) to 60.5% (31–60 days).
- Distance to the clinic materially affects attendance — 68.1% no-show rate at 30–50km vs 46.5% within 5km.
- Past behaviour predicts future behaviour — patients with 3+ prior no-shows miss 68.8% of appointments vs 43.5% for a clean record.
- Reminders help modestly and SMS outperforms Email and WhatsApp.
- Waiting time, appointment type, time of day, day of week, and age group show no meaningful relationship with no-shows in this dataset.

Recommendations

- Prioritise SMS as the default reminder channel, using WhatsApp/Email as secondary channels.
- Prioritise proactive follow-up for appointments booked far in advance.
- Prioritise patients with previous no-show history for reminder and administrative support.
- Test reminder timing and channels through controlled monitoring; this analysis shows association, not
- causation.
- Investigate access/travel support options for patients living farther from the clinic, subject to approved clinic
- procedures.
- Use appointment type and day patterns to focus operational resources.
- Maintain data-quality monitoring for missing distance, waiting-time and reminder-channel information.

Limitations

- The dataset is fictional/synthetic; findings should be validated against real operational data before acting on them.
- This is a descriptive/associational analysis, not a controlled experiment or predictive model — the drivers found are correlations, not proven causal effects.
- distance_to_clinic_km (1.8%) and waiting_time_minutes (1.2%) have small amounts of missing data, excluded row-wise rather than imputed.
- Cancelled appointments (5.3%) were treated as a separate outcome from No-Shows; a different treatment could shift rates slightly.

Part 7 — Cross-Track Collaboration

Track interacted with: Data Science.

Information exchanged: the three strongest no-show predictors (booking_lead_days, distance_to_clinic_km, previous_no_shows), the KPI definitions, and the high-risk patient segment (2+ prior no-shows, 10.6% of appointments, 61.0% no-show rate).

Why relevant: these findings directly inform Data Science's Week 5 feature engineering and baseline model, giving a validated starting point for candidate features.

What changed/improved: Data Science can prioritise these three fields as first-pass model features and use the high-risk segment definition to sanity-check model outputs.

Part 8 — Week 5 Project Summary

1. Planned for Week 5: data preparation, EDA, KPI development, an initial dashboard, and business insights, building on the Week 4 foundation.

2. Completed: full data-quality assessment, EDA across six dimensions, 5 defined/calculated KPIs, a consolidated dashboard, and 6 business insights with recommendations.

3. Key findings: booking lead time, distance, and prior no-show history are the clearest drivers of missed appointments; reminders help modestly (SMS best); scheduling factors show little relationship with attendance.

4. Major challenges: none from a data-quality perspective — the dataset was clean and internally consistent.

5. Key decisions: treated Cancelled as distinct from No-Show throughout.

6. Changes to Week 4 approach: none required — the Week 4 KPI shortlist held up against the real data.

7. Cross-track collaboration completed: shared top predictive features and the high-risk segment with the Data Science track.

8. Remaining work: statistical significance testing, and validation once a Data Science baseline model is available.

9. Proposed Week 6 focus: a stakeholder-ready Power BI/Tableau dashboard, monitoring model validation of these drivers, and scoping a reminder-strategy test.